

DD_SIP-05 - Promotional Campaigns

Version: 1.0 Status: Draft for review Bundle: 09_catalogs Audience: commercial developers, sales developers, billing developers, frontend developers, QA Related DDs:

- DD_SIP-03 - Discount Assignment
- DD_SIP-04 - Bundle Launch
- DD_PLM-CFG-04 - Discount Catalog
- DD_SALES-01 - Lead, Territory, and Franchise Sales Management
- DD_EM-01 - Franchise Management
- DD_EM-03 - CVM and Retention
- DD_FUL-02 - Order Capture
- DD_BIL-01 - Charging Engine Core
- DD_EM-CFG-04 - Approval Workflow Catalog
- FOUNDATION_DROOLS - Rules Engine Foundation
- FE-APP-01 - BSS Backoffice Web App
- FE-APP-02 - Outdoor Sales Franchise App
- FE-APP-04 - Customer Self Care Frontoffice App

MVP simplification baseline: first release supports simple campaign eligibility, lifecycle approval, and binding to discount assignment. Do not build a full marketing automation suite, journey builder, or advanced experimentation platform in v1.

1. Purpose

SIP-05 defines promotional campaign lifecycle, eligibility, targeting, channel governance, and campaign-driven discount assignment.

A campaign is a time-bound commercial program with a goal, target population, offer terms, channels, limits, approvals, and reporting attribution.

Examples:

- acquisition campaign for new customers in Nairobi franchise territories;
- retention campaign for customers at churn risk;
- upsell campaign from Fiber 50M to Fiber 100M;
- win-back campaign for recently terminated customers;
- recovery campaign for customers suspended due to non-payment;
- seasonal promotion with free installation discount;
- campaign limited to door-to-door sales app and selected franchises.

SIP-05 does not define discount amounts; PLM-CFG-04 does. SIP-05 does not create invoice discounts directly; SIP-03 creates assignment records and DIS-OP/BIL applies them at runtime.

2. Ownership Boundary

Concern	Owner
Campaign lifecycle, targeting, channel eligibility, budget/limits	SIP-05
Discount definition	PLM-CFG-04
Discount assignment created from campaign	SIP-03
Bundle offer used by campaign	SIP-04
Leads, territories, sales activities, conversion attribution	SALES-01

Concern	Owner
Franchise master/scope	EM-01
Customer retention scoring and CVM segments	EM-03 / REP inputs
Configurable campaign approval process	EM-CFG-04
Order capture	FUL-02
Billing-time application	DIS-OP-01 / BIL-01
Campaign eligibility and targeting rules	Drools rules engine where policy is operator-variable

Rule: SIP-05 decides whether a customer/lead/order is eligible for a promotion and which assignment request should be created. It does not apply money to invoices or wallets.

Approval rule: SIP-05 does not define approval steps locally. Campaign launch approval should call EM-CFG-04 for action codes such as CAMPAIGN_LAUNCH_APPROVAL. Camunda is used only when the selected EM-CFG-04 policy is BPMN-backed.

3. Campaign Types

Campaign type	Use
ACQUISITION	New customer acquisition, often via sales app/franchise.
RETENTION	Keep customer from leaving; often care/backoffice driven.
UPSELL	Move customer to higher package/bundle.
CROSS_SELL	Add service or bundle component.
WINBACK	Target recently terminated/lost customers.
RECOVERY	Recover suspended/non-paying customer with approved terms.
LOYALTY	Reward long-term customer.
STAFF_PARTNER	Staff, partner, or special group offer.
REGIONAL	Region/franchise/territory-specific campaign.

4. High Level Flow

```

flowchart TD
    A["Commercial user creates campaign draft"]
    B["Define target rules, channel, dates, limits"]
    C["Attach discount/bundle/offer terms"]
    D["Validate against catalog, franchise, package, budget"]
    E{"Approval required?"}
    F["Approval review"]
    G["Activate campaign"]
    H["Sales/backoffice/self-care checks eligibility"]
    I{"Eligible?"}
    J["Create campaign participation"]
    K["Request SIP-03 discount assignment"]
    L["Order/subscription flow proceeds"]
    M["Track conversion and usage"]

    A --> B --> C --> D --> E
    E -->|Yes| F --> G
    E -->|No| G
    G --> H --> I
    I -->|No| M
    I -->|Yes| J --> K --> L --> M

```

5. Campaign Lifecycle

Status	Meaning	Allowed next statuses
DRAFT	Campaign is being configured.	READY_FOR_REVIEW, CANCELLED
READY_FOR_REVIEW	Validation passed, waiting approval.	APPROVED, REJECTED, DRAFT
APPROVED	Approved but not active until start date.	ACTIVE, CANCELLED

Status	Meaning	Allowed next statuses
ACTIVE	Eligibility can return campaign as available.	PAUSED, ENDED
PAUSED	Temporarily hidden from new eligibility.	ACTIVE, ENDED
ENDED	Campaign finished. No new participation.	none
REJECTED	Approval rejected.	DRAFT
CANCELLED	Cancelled before/without launch.	none

6. Tables

6.1 promotion_campaign

Stores campaign master data.

Use this table when commercial/marketing defines a campaign program.

Column	Type	Purpose	Sample
campaign_id	UUID PK	Campaign id.	camp-001
operator_code	TEXT	Operator/tenant.	WIK
campaign_code	TEXT	Stable unique code.	ACQ_NRB_FREE_INSTALL_JULY
display_name	TEXT	UI/customer label.	Free Installation July
campaign_type	TEXT	Type from section 3.	ACQUISITION
status	TEXT	Lifecycle status.	ACTIVE
start_date	DATE	First eligible date.	2026-07-01
end_date	DATE	Last eligible date.	2026-07-31
owner_team_code	TEXT	Commercial owner.	COMMERCIAL_ACQ
budget_limit_amount	DECIMAL(18,2) NULL	Optional budget cap.	500000.00
currency_code	TEXT	Currency.	KES
max_participants	INT NULL	Optional cap.	1000
created_by_user_id	TEXT	Creator.	usr-commercial-001
created_at	TIMESTAMP	Creation timestamp.	2026-06-15T09:00:00Z
updated_at	TIMESTAMP	Last update.	2026-06-15T09:30:00Z

Recommended indexes:

- Unique (operator_code, campaign_code).
- (operator_code, status, start_date, end_date).

6.2 promotion_campaign_offer

Stores what the campaign offers.

Use this table to attach discount codes, bundle codes, package refs, or service terms to the campaign.

Column	Type	Purpose	Sample
offer_id	UUID PK	Offer id.	pco-001
campaign_id	UUID	Campaign id.	camp-001
offer_type	TEXT	DISCOUNT, BUNDLE, PACKAGE, MESSAGE_ONLY.	DISCOUNT
discount_code	TEXT NULL	PLM-CFG-04 discount.	ACQ_FREE_INSTALL
bundle_code	TEXT NULL	SIP-04 bundle.	FIBER_TV_STARTER
package_ref	TEXT NULL	SIP-01 package.	pkg-fiber-100m
assignment_scope_type	TEXT NULL	SIP-03 scope if discount.	ORDER
validity_mode	TEXT	CAMPAIGN_WINDOW, DURATION_FROM_ASSIGNMENT, ORDER_ONLY.	ORDER_ONLY
duration_days	INT NULL	Duration if dynamic.	90

Column	Type	Purpose	Sample
priority	INT	Offer priority.	100
status	TEXT	ACTIVE, SUSPENDED, RETIRED.	ACTIVE

Recommended indexes:

- (campaign_id, status).
- (discount_code).
- (bundle_code).

6.3 promotion_campaign_target_rule

Stores campaign targeting rules.

Use this table to define who is eligible. Rules are evaluated by the campaign eligibility API.

Column	Type	Purpose	Sample
target_rule_id	UUID PK	Rule id.	pct-001
campaign_id	UUID	Campaign id.	camp-001
rule_type	TEXT	CUSTOMER_SEGMENT, PACKAGE, REGION, FRANCHISE, CHANNEL, TENURE, PAYMENT_STATUS, LEAD_STATUS, HOME_PASS, CUSTOM.	FRANCHISE
operator	TEXT	IN, NOT_IN, EQ, GTE, LTE, BETWEEN.	IN
rule_value_json	JSONB	Rule value.	{"franchiseIds":["fr_01J_NRB_002"]}
hard_exclusion	BOOLEAN	If true, failed rule blocks eligibility.	true
display_order	INT	Evaluation/display order.	10
status	TEXT	ACTIVE, RETIRED.	ACTIVE

Recommended indexes:

- (campaign_id, status, display_order).
- (rule_type, status).

6.4 promotion_campaign_channel

Stores channels allowed to present or redeem campaign.

Use this table so a promotion can be Backoffice-only, Sales-app-only, Self-care-only, USSD-only, or multi-channel.

Column	Type	Purpose	Sample
channel_id	UUID PK	Channel row id.	pcc-001
campaign_id	UUID	Campaign id.	camp-001
channel_code	TEXT	BACKOFFICE, SALES_APP, SELF_CARE, USSD, PARTNER_API.	SALES_APP
requires_agent_attribution	BOOLEAN	Agent/franchise must be captured.	true
status	TEXT	ACTIVE, SUSPENDED, RETIRED.	ACTIVE

Recommended indexes:

- (campaign_id, channel_code, status).

6.5 promotion_campaign_participation

Stores one eligible/redeemed participation for a lead, order, customer, or subscription.

Use this table when a campaign is selected or redeemed. It prevents duplicate redemption and supports conversion reporting.

Column	Type	Purpose	Sample
participation_id	UUID PK	Participation id.	pcp-001
campaign_id	UUID	Campaign id.	camp-001

Column	Type	Purpose	Sample
operator_code	TEXT	Operator/tenant.	WIK
participant_type	TEXT	LEAD, ORDER, CUSTOMER, ACCOUNT, SUBSCRIPTION.	ORDER
participant_ref_id	TEXT	Participant id.	ORD-1001
customer_id	TEXT NULL	Customer id.	CUS-100045
lead_id	TEXT NULL	Lead id.	lead-001
order_id	TEXT NULL	Order id.	ORD-1001
franchise_id	TEXT NULL	Franchise attribution.	fr_01J_NRB_002
agent_user_id	TEXT NULL	Sales/care agent.	usr-sales-003
status	TEXT	ELIGIBLE, SELECTED, REDEEMED, REJECTED, CANCELLED.	REDEEMED
assignment_id	TEXT NULL	SIP-03 discount assignment id.	da-101
created_at	TIMESTAMP	Creation timestamp.	2026-07-02T10:00:00Z
redeemed_at	TIMESTAMP NULL	Redemption timestamp.	2026-07-02T10:05:00Z

Recommended indexes:

- Unique (campaign_id, participant_type, participant_ref_id).
- (campaign_id, status).
- (customer_id, status).
- (franchise_id, status).

6.6 promotion_campaign_approval

Stores the SIP-05 side reference to a shared EM-CFG-04 approval request.

Use this table when campaign launch requires commercial, finance, legal, or management approval. The configurable policy, steps, approver routing, BPMN process key, user tasks, and final decision are owned by EM-CFG-04.

Column	Type	Purpose	Sample
approval_id	UUID PK	Approval id.	pca-001
campaign_id	UUID	Campaign id.	camp-001
approval_policy_code	TEXT	EM-CFG-04 policy code.	CAMPAIGN_COMMERCIAL_FINANCE
approval_request_id	TEXT NULL	EM-CFG-04 runtime approval request id.	apr-camp-001
approval_mode	TEXT	AUTO, SINGLE_STEP, MULTI_STEP, BPMN.	BPMN
camunda_process_instance_id	TEXT NULL	Camunda instance id when BPMN-backed.	cam-proc-camp-001
status	TEXT	PENDING, APPROVED, REJECTED, CANCELLED.	APPROVED
assigned_queue_code	TEXT	Approval queue.	COMMERCIAL_APPROVERS
actor_user_id	TEXT NULL	Approver.	usr-finance-001
decision_comment	TEXT NULL	Decision note.	Approved budget and campaign terms.
created_at	TIMESTAMP	Creation timestamp.	2026-06-15T09:40:00Z
decided_at	TIMESTAMP NULL	Decision timestamp.	2026-06-15T11:10:00Z

Recommended indexes:

- (campaign_id, status).
- (assigned_queue_code, status).

7. APIs

7.1 Create Campaign

POST /api/promotional-campaigns
Idempotency-Key: campaign-acq-nrb-free-install-july
Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "campaignCode": "ACQ_NRB_FREE_INSTALL_JULY",
  "displayName": "Free Installation July",
  "campaignType": "ACQUISITION",
  "startDate": "2026-07-01",
  "endDate": "2026-07-31",
  "ownerTeamCode": "COMMERCIAL_ACQ",
  "budgetLimitAmount": 500000,
  "currencyCode": "KES",
  "maxParticipants": 1000
}
```

7.2 Add Offer

POST /api/promotional-campaigns/{campaign_id}/offers
Content-Type: application/json

```
{
  "offerType": "DISCOUNT",
  "discountCode": "ACQ_FREE_INSTALL",
  "assignmentScopeType": "ORDER",
  "validityMode": "ORDER_ONLY",
  "priority": 100
}
```

7.3 Add Target Rule

POST /api/promotional-campaigns/{campaign_id}/target-rules
Content-Type: application/json

```
{
  "ruleType": "FRANCHISE",
  "operator": "IN",
  "ruleValue": {
    "franchiseIds": ["fr_01J_NRB_002"]
  },
  "hardExclusion": true
}
```

7.4 Validate Campaign

POST /api/promotional-campaigns/{campaign_id}/validate

Validation checks dates, offers, discount/bundle refs, channel config, target rules, budget caps, and approval policy.

7.5 Approve And Activate

POST /api/promotional-campaigns/{campaign_id}/approve
POST /api/promotional-campaigns/{campaign_id}/activate

Activation is allowed only after approval if approval policy requires it and start_date <= today <= end_date.

7.6 Eligibility Check

Used by Sales App, Backoffice, Self-care, USSD, and order capture.

POST /api/promotional-campaigns/eligibility
Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "channelCode": "SALES_APP",
  "leadId": "lead-001",
  "orderId": "ORD-1001",
  "customerId": "CUS-100045",
  "packageRef": "pkg-fiber-100m",
  "bundleCode": "FIBER_TV_STARTER",
  "franchiseId": "fr_01J_NRB_002",
  "homepassId": "HP-001",
  "onDate": "2026-07-02"
}
```

Response:

```
{
  "eligibleCampaigns": [
    {
      "campaignId": "camp-001",
      "campaignCode": "ACQ_NRB_FREE_INSTALL_JULY",
      "displayName": "Free Installation July",
      "offers": [
        {
          "offerType": "DISCOUNT",
          "discountCode": "ACQ_FREE_INSTALL",
          "assignmentScopeType": "ORDER"
        }
      ]
    }
  ],
  "rejectedCampaigns": [
    {
      "campaignCode": "UPSELL_200M",
      "reasonCode": "PACKAGE_NOT_ELIGIBLE"
    }
  ]
}
```

7.7 Redeem Campaign

POST /api/promotional-campaigns/{campaign_id}/redeem
 Idempotency-Key: redeem-camp-001-order-ORD-1001
 Content-Type: application/json

```
{
  "participantType": "ORDER",
  "participantRefId": "ORD-1001",
  "customerId": "CUS-100045",
  "leadId": "lead-001",
  "orderId": "ORD-1001",
  "franchiseId": "fr_01J_NRB_002",
  "agentUserId": "usr-sales-003",
  "selectedOfferIds": ["pco-001"]
}
```

Redeem creates participation and calls SIP-03 for discount assignment when offer type is DISCOUNT.

8. Events

Event	Topic	When emitted
PromotionalCampaignCreated	sophix.commercial.campaign.created	Campaign draft created.
PromotionalCampaignApproved	sophix.commercial.campaign.approved	Campaign approved.
PromotionalCampaignActivated	sophix.commercial.campaign.activated	Campaign becomes active.
PromotionalCampaignPaused	sophix.commercial.campaign.paused	Campaign paused.
PromotionalCampaignEnded	sophix.commercial.campaign.ended	Campaign ended.
CampaignParticipationCreated	sophix.commercial.campaign.participationcreated	Lead/order/customer selected campaign.
CampaignRedeemed	sophix.commercial.campaign.redeemed	Campaign redeemed and assignments requested.

9. Business Rules

Rule	Description	Reason
R-SIP-CAMP-01	Campaign code must be unique per operator.	Stable reporting and API ref.
R-SIP-CAMP-02	Campaign must have valid start/end dates.	Time-bound governance.
R-SIP-CAMP-03	Active campaign must have at least one active offer or message-only purpose.	Avoid empty campaigns.
R-SIP-CAMP-04	Discount offers must reference active PLM-CFG-04 discounts.	Avoid invalid redemption.
R-SIP-CAMP-05	Bundle offers must reference active SIP-04 bundles.	Avoid unsellable bundle campaigns.
R-SIP-CAMP-06	Eligibility must evaluate channel, region/franchise, package/bundle, customer/lead state, and date.	Correct targeting.

Rule	Description	Reason
R-SIP-CAMP-07	Participation must be idempotent per campaign and participant.	Avoid duplicate redemption.
R-SIP-CAMP-08	Campaign budget and max participant caps must block new redemptions when reached.	Commercial control.
R-SIP-CAMP-09	Redeeming a discount campaign must create SIP-03 assignment; it must not apply invoice/wallet impact directly.	Ownership boundary.
R-SIP-CAMP-10	Sales app can show only campaigns allowed for the agent's channel/franchise/territory.	Field governance.
R-SIP-CAMP-11	Campaign approval policy and tasks must be handled through EM-CFG-04.	Shared approval governance.

10. Rule Engine Guidance

Use Drools for campaign eligibility because campaign targeting changes often and combines many facts.

Rule file	Purpose
campaign-eligibility.drl	Evaluate campaign target rules by channel, franchise, territory, package, bundle, lead/customer state, payment state, date, and caps.
campaign-redemption.drl	Decide whether redemption is allowed and which offer should create SIP-03 assignments.
campaign-approval.drl	Decide whether campaign launch requires EM-CFG-04 approval based on budget, type, channel, and discount value.

Fixed checks such as campaign code uniqueness, date parsing, and idempotency remain application validation.

Example rule:

```
rule "Sales acquisition campaign requires matching franchise and package"
when
    $c : CampaignEligibilityFact(campaignType == "ACQUISITION", channelCode == "SALES_APP")
    $ctx : CampaignContextFact(franchiseAllowed == true, packageAllowed == true, capReached == false)
    $out : CampaignEligibilityDecision()
then
    $out.addEligibleCampaign($c.getCampaignCode());
end
```

11. Frontend Usage

11.1 FE-APP-01 Backoffice

Backoffice must support:

- campaign list and status filters;
- campaign create/edit;
- offer setup;
- target rule setup;
- channel setup;
- validation and approval;
- activate/pause/end;
- participation/redeem reporting;
- drill-through to SIP-03 assignments and SALES leads/orders.

11.2 FE-APP-02 Sales App

Sales app must support:

- retrieve eligible campaigns for a lead/HomePass/package/bundle;

- show campaign terms and expiry date;
- include selected campaign code in order conversion;
- redeem campaign online only;
- cache campaign summaries for display, but eligibility/redeem must be server-confirmed.

11.3 FE-APP-04 Self-Care

Self-care may show campaigns only when:

- channel SELF_CARE is active;
- customer eligibility passes;
- terms are customer-safe;
- redemption can be completed through supported self-care workflows.

12. Infra Recommendation

For a medium budget deployment, run SIP-05 inside the commercial/catalog API deployment. Do not create a separate campaign service initially.

Component	Recommendation
API	Existing commercial/catalog API pods expose <code>/api/promotional-campaigns</code> .
DB	Same PostgreSQL cluster/schema family as commercial catalog.
Workers	Existing worker pool handles activation/end, cap sync, cache refresh.
Events	Kafka/MSK topics under <code>sophix.commercial.campaign.*</code> .
Cache	Read model/cache for active campaigns by channel/franchise/region.

AWS deployment: use existing ECS/EKS service and RDS cluster. Add CloudWatch alarms for failed activations, eligibility API latency, redemption duplicate spikes, cap overrun attempts, and SIP-03 assignment failures from campaign redemption.

13. Testing Requirements

13.1 Unit Tests

- Campaign date validation.
- Offer validation against discount/bundle catalog.
- Target rule evaluation.
- Channel filtering.
- Budget/max participant caps.
- Duplicate participation prevention.
- Drools campaign eligibility/redemption/approval decisions.

13.2 Integration Tests

- Eligibility reads SALES/EM/RLM/SIP context through APIs.
- Redeem creates campaign participation.
- Discount campaign redemption calls SIP-03 assignment API.
- Sales app receives only channel/franchise-eligible campaigns.
- Paused/ended campaign disappears from eligibility.
- EM-CFG-04 campaign approval request/callback activates campaign when required.

13.3 End To End Tests

- Sales agent sees acquisition campaign, converts lead, campaign creates order discount assignment.
- Retention campaign visible in Backoffice but not Sales App.
- Campaign cap reached blocks new redemption.
- Campaign pause blocks new participation but keeps existing participation history.

14. Definition Of Done

- Campaign tables and APIs exist.
- Campaign lifecycle, approval, validation, activation, pause, and end are implemented.
- EM-CFG-04 is used for configurable campaign approval policies.
- Drools rule files exist for configurable eligibility, redemption, and approval decisions.
- Eligibility API evaluates channel/franchise/package/customer/date.
- Redeem API creates participation and calls SIP-03 for assignment when needed.
- Backoffice can configure and monitor campaigns.
- Sales app can display and redeem eligible campaigns.
- Events are emitted for reporting and cache refresh.
- Tests cover acquisition, retention, channel restrictions, caps, and redemption.